Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/23/2013
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism in Micro- and Macro Scales		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	09/30/2012
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Change in end date to 9/30/2012 per PI (Ed., 2/28/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Didion, Jeffrey (NASA Goddard Space Flight Center) Cassel, Kevin (Illinois Institute of Technology)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably and for example reaching much above 300 W/cm2 for refrigerant R-123. EDITOR's NOTE: Grant continues as "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism"; grant NNX12AR32G, effective 10/1/2012.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Innovative technology is being developed partially via this grant from NASA. Specifically, a two-phase heat transport device is being developed in meso- and micro-scale, driven with electric conduction mechanism. Two setups have been designed and fabricated based on the above mentioned concept. The first one, developed at PI's lab at IIT (to be moved to WPI in May 2012). The second one, developed as a collaborative effort between PI and Co-PI, and is located at Co-PI's lab at NASA-GSFC. The PI and Co-PI's teams are scheduled to fly on NASA's 0-g flight in April 2012 to examine this technology. This unique two-phase heat transport device actively transports heat compared to the current heat pipe technology that is a passive system.
Task Progress:	Electrohydrodynamic (EHD) pumping methods rely on the interaction between electric fields and the flow fields of a dielectric fluid. Conduction pumping is one EHD pumping mechanism, which offers many advantages over other EHD pumping methods (for example, a simple design, no degradation of the working fluid, and no need for a temperature gradient). Conduction pumping can be used in an adiabatic context to pump a working fluid or it can be applied to deliver substantial enhancement to single- and two-phase heat transfer processes. Experimental studies of conduction pumping to date have focused on macro-scale devices with applied voltages on the order of 10 kV. However, like many other EHD concepts, conduction pumping depends primarily on the intensity of the imposed electric field. Therefore, at the micro-scale, the reduced physical size can be accompanied by a reduction in the magnitude of the applied voltage to levels that are significantly more manageable. Furthermore, the simplicity of EHD conduction pumps, such as the lack of moving parts, high reliability, and physical compactness, make them an attractive method for pressure generation in micro-scale fluid and heat transfer devices. This experimental study examines the fundamental behavior and performance of EHD conduction pumping at the micro-scale. The pump is embedded in a rectangular, adiabatic micro-channel, and the single-phase flow and pressure generation of the pump are characterized. The EHD pumping of single- and two-phase flows in micro-channels in the presence of heat transfer is also studied and the corresponding enhancement to single- and two-phase heat transfer is quantified. In an additional study, micro-scale electrodes are embedded within a flat, heated surface to examine the ability of the conduction pump to provide electrically-enhanced wetting of the heated surface during pool boiling. A model based on hydrodynamic instability theory is generated to quantify the influence of the EHD conduction pumping on the pool boiling critical heat flux. Finally, this technology is incorporated into two unique, novel, heat transport devices akin to a heat pipe but in which the primary driving force for the liquid is conduction pumping, not capillarity. Research on heat transport in micro-scale has been generating much interest in recent years due to the development of state-of-the-art, high-powered electronics used in aerospace, outerspace, and terrestrial applications and the large amount of heat produced during their operation. Micro-scale, two-phase flow heat transport devices are seen as one solution to this problem of high heat-flux removal. Micro-scale devices have extremely high heat fluxes due to the small heat transfer surface area. In addition, the need for robust, non-mechanical, lightweight, low-noise and low-vibration devices in specialized aerospace applications has led researchers to investigate electrically driven flow devices rather than their mechanical counterparts. This report for the first time presents the results of an experimental study of a unique micro-scale heat transport device that is driven by electrohydrodynamic (EHD) conduction pumping. Results from ground-based single-phase experiments with a micro-scale EHD pump are compared to experiments conducted on board a variable-gravity parabolic flight. Data show that the EHD pump functions well in both environments and can potentially be used in heat transport devices in the absence of gravity.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Peer-reviewed Journals	Pearson MR, Seyed-Yagoobi J. "Experimental study of EHD conduction pumping at the meso- and micro-scale." Journal of Electrostatics, 2011 Dec;69(6):479-85. http://dx.doi.org/10.1016/j.elstat.2011.06.003 , Dec-2011